

The Algorithm Is Not the Answer

A Methodological Review of Metaheuristic Controller Tuning

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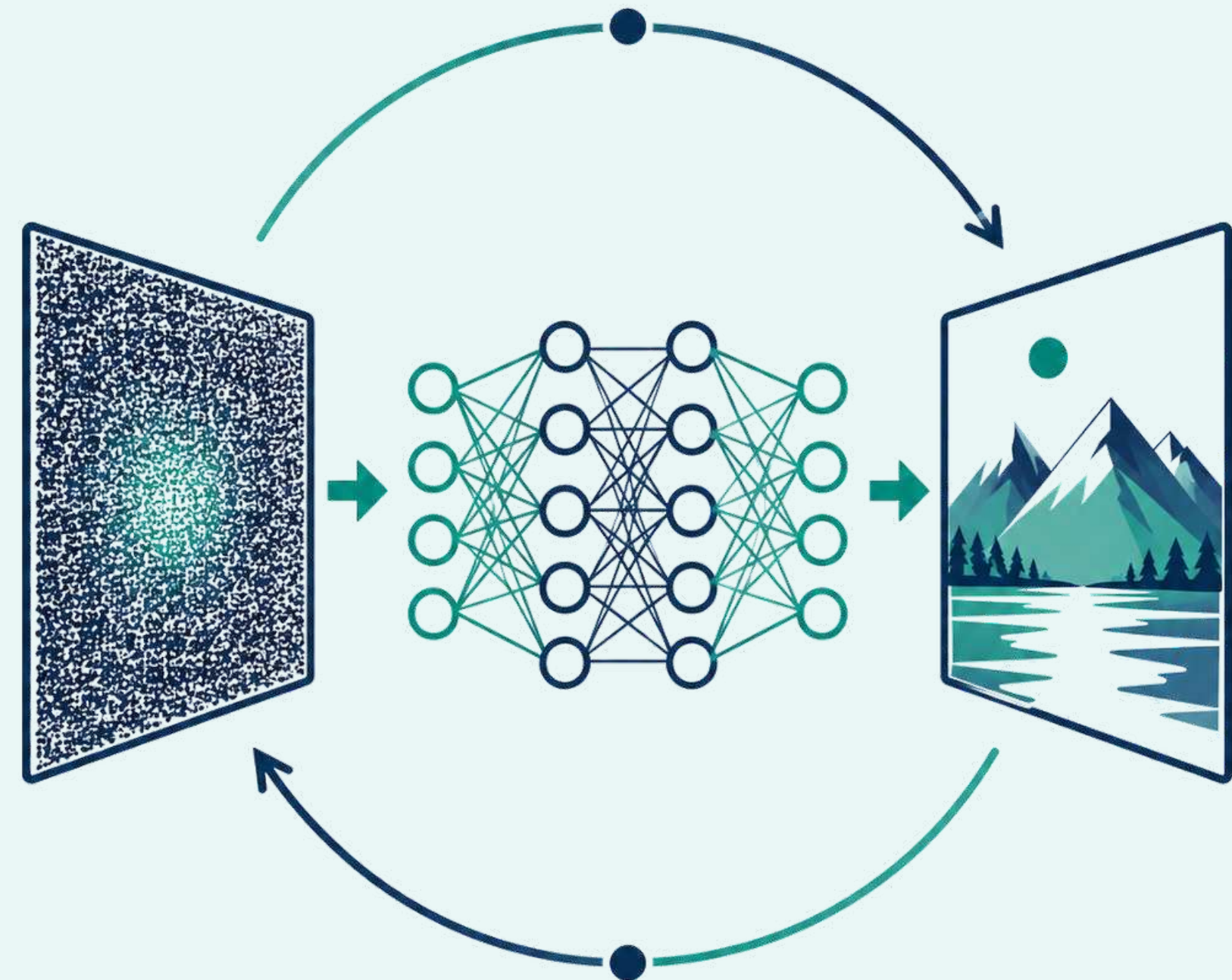
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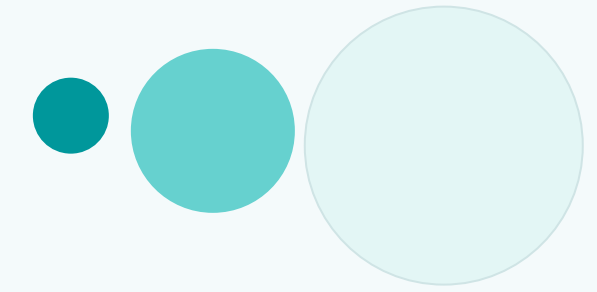


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Controller tuning is a formulation problem - not merely an algorithm race.

Introduction

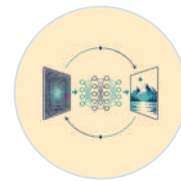


Why metaheuristic tuning matters



Modern systems are harder to tune

Nonlinearity, coupling, and constraints challenge classical tuning rules.



Metaheuristics are attractive

Gradient-free global search can navigate non-convex, expensive, and constrained tuning spaces.



But novelty is not enough

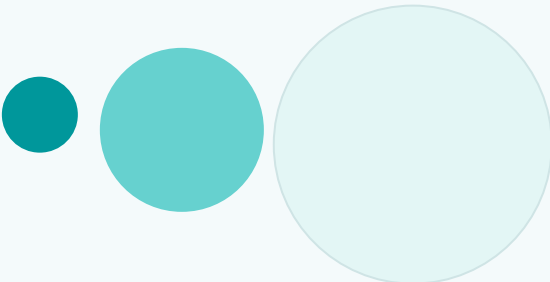
Weak objectives, unfair baselines, and simulation-only evidence can inflate performance claims.

Core argument

Credible progress requires problem-structure-aware optimization - not another algorithm name.

The review evaluates how tuning is formulated, benchmarked, validated, and reported.

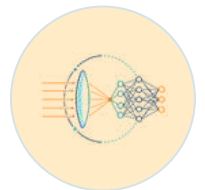
Review Scope and Analytical Lens



Review scope



Studies published in 2025-2026.



Five engineering application domains.



Critical analysis of rigor and transferability.

Analytical lens



Controller families and tuning dimensionality.



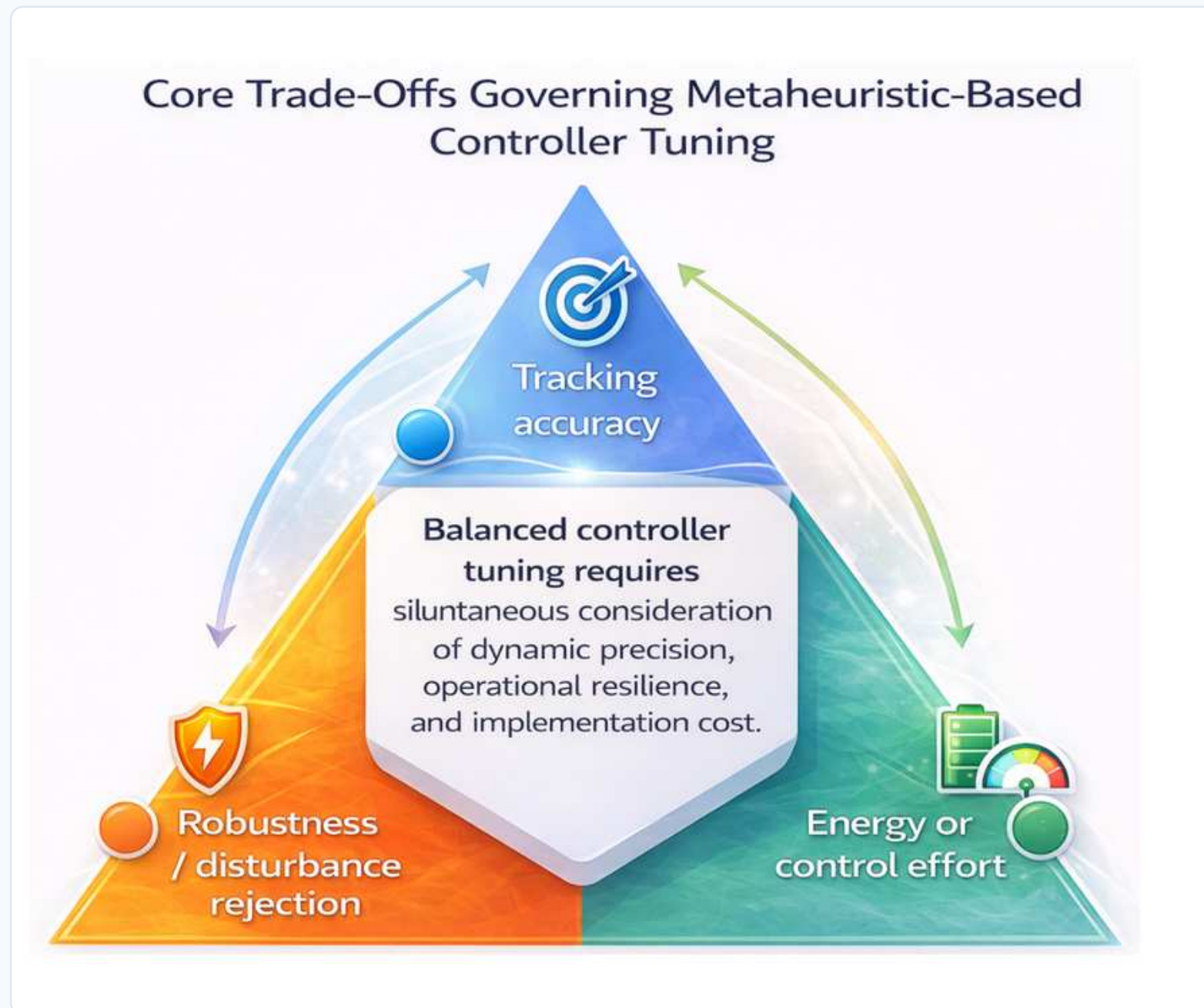
Optimizer families and search mechanisms.



Objective families and engineering meaning.

Central question: how should tuning reflect real engineering priorities and constraints?

Controller tuning as a trade-off problem



Paper Fig. 1: tracking-robustness-effort trade-off

Offline

Parameters optimized before deployment; most common in the reviewed literature.

Online

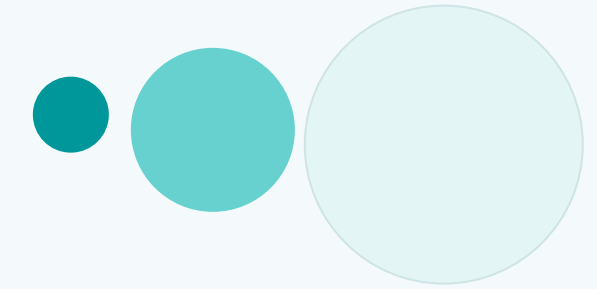
Parameters updated during operation; suited to uncertain environments.

Hybrid

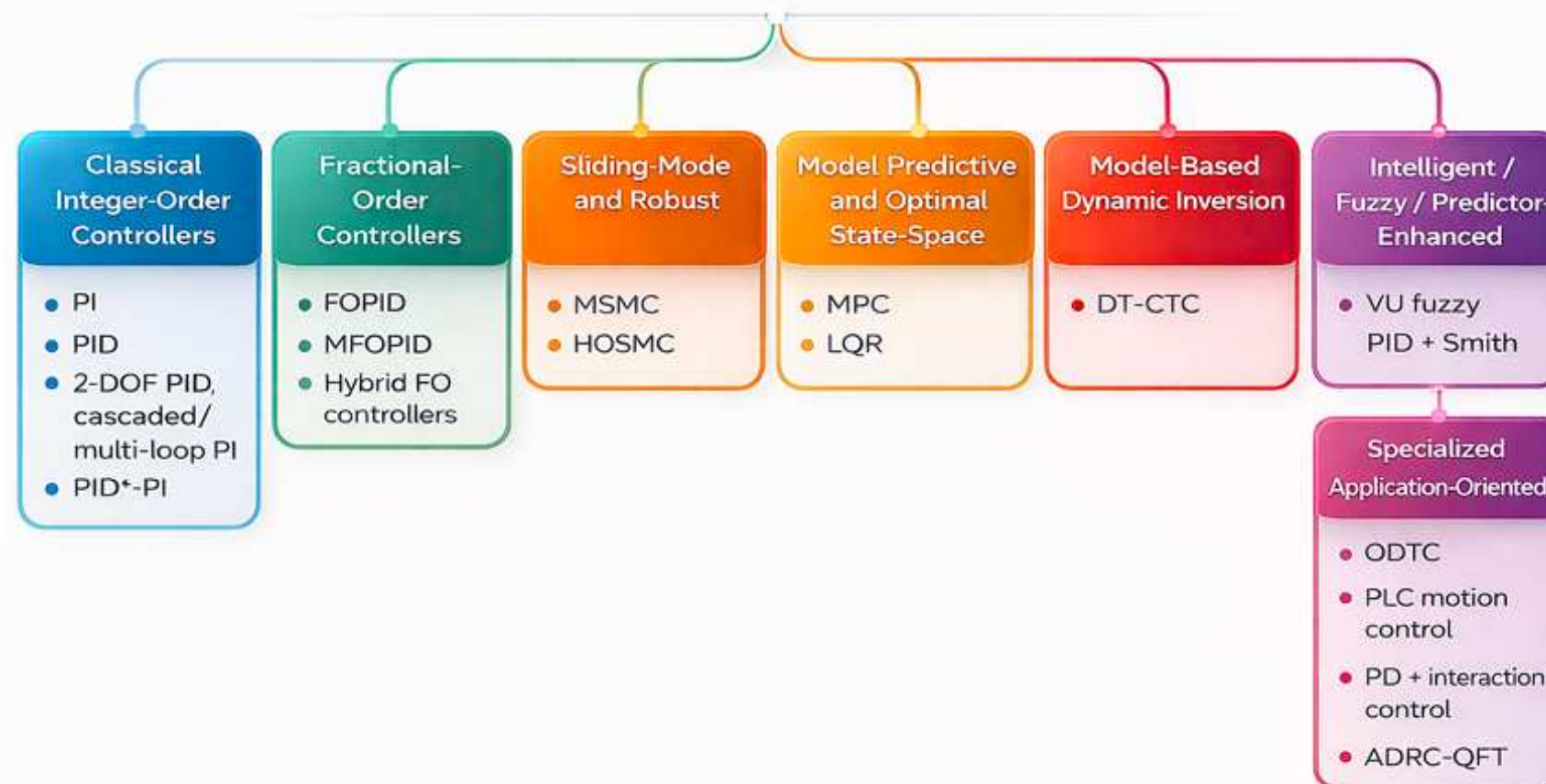
Offline initialization plus online refinement; flexible but methodologically underdeveloped.

Critical gap: Systematic guidance for selecting the tuning paradigm remains limited.

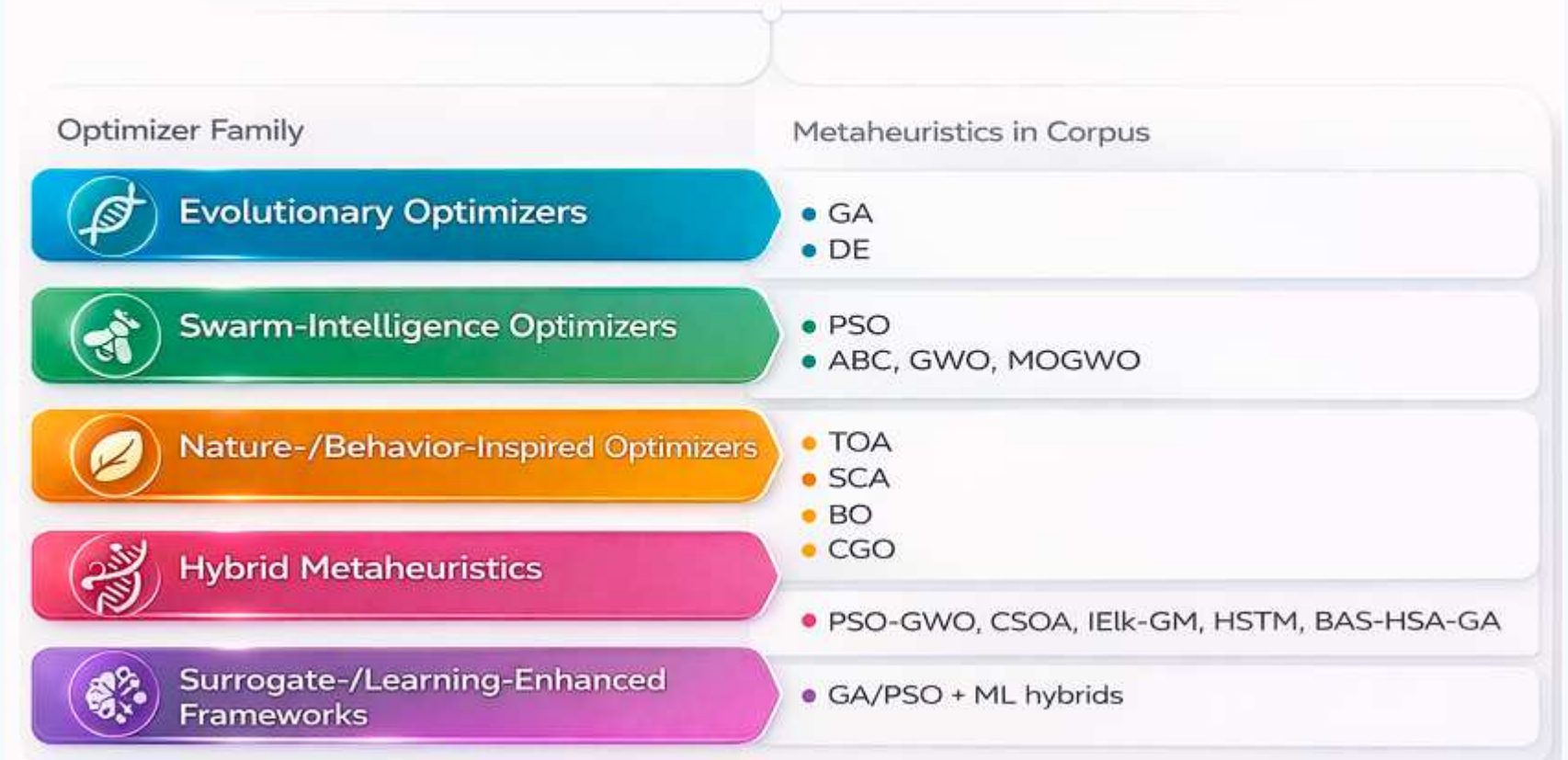
Three Coupled Choices Shape Outcomes



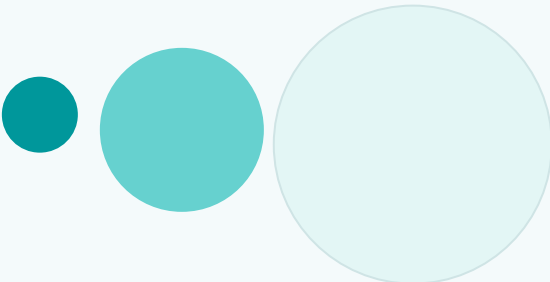
Controller Types Identified in the Reviewed Corpus



Optimizer Families Identified in the Reviewed Corpus



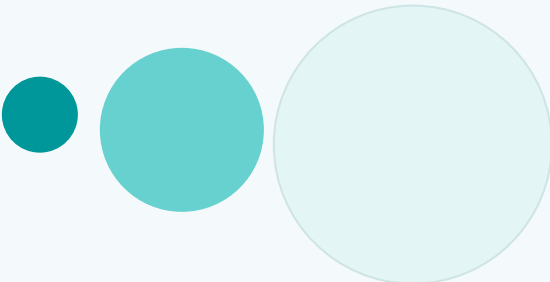
Objective Design Determines Physical Meaning



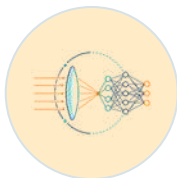
OBJECTIVE FUNCTION FAMILIES IN THE REVIEWED CORPUS

Family	Representative Forms	Optimization Focus	Representative Refs.
Classical error-integral	ISE, IAE, IE, MSE	Direct error minimization	[6], [5], [2], [9]
Time-weighted error	ITAE, ITSE, ISTSE	Penalize persistent error	[1], [21], [3], [7], [10], [14], [18]
Composite time-domain	error + settling time; error + effort; error + smoothness	Multi-criterion transient shaping	[20], [11], [4], [15], [16], [8], [19]
Energy-/physics-aware	torque ripple, losses, efficiency, energy consumption	Physical performance optimization	[20], [17], [15]
Structured predictive/optimal	MPC multi-term cost; Pareto-based weighting	Internal cost-structure tuning	[13], [14]
Robust/frequency-domain	min-max; penalty-based robust cost; frequency-domain objectives	Robustness-oriented design	[2], [10], [12]

Metaheuristic Value Is Application-Dependent



Coupling, nonlinearity, high dimension, and delay



EV charging & drives

Cross-stage coordination, transient reversal, THD, torque ripple, losses, and efficiency.



Power systems

Frequency restoration and tie-line control under renewables, delays, and attacks.



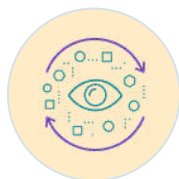
Robotics & motion

Nonlinear, high-dimensional tracking with smoothness, effort, and safety trade-offs.



Process control

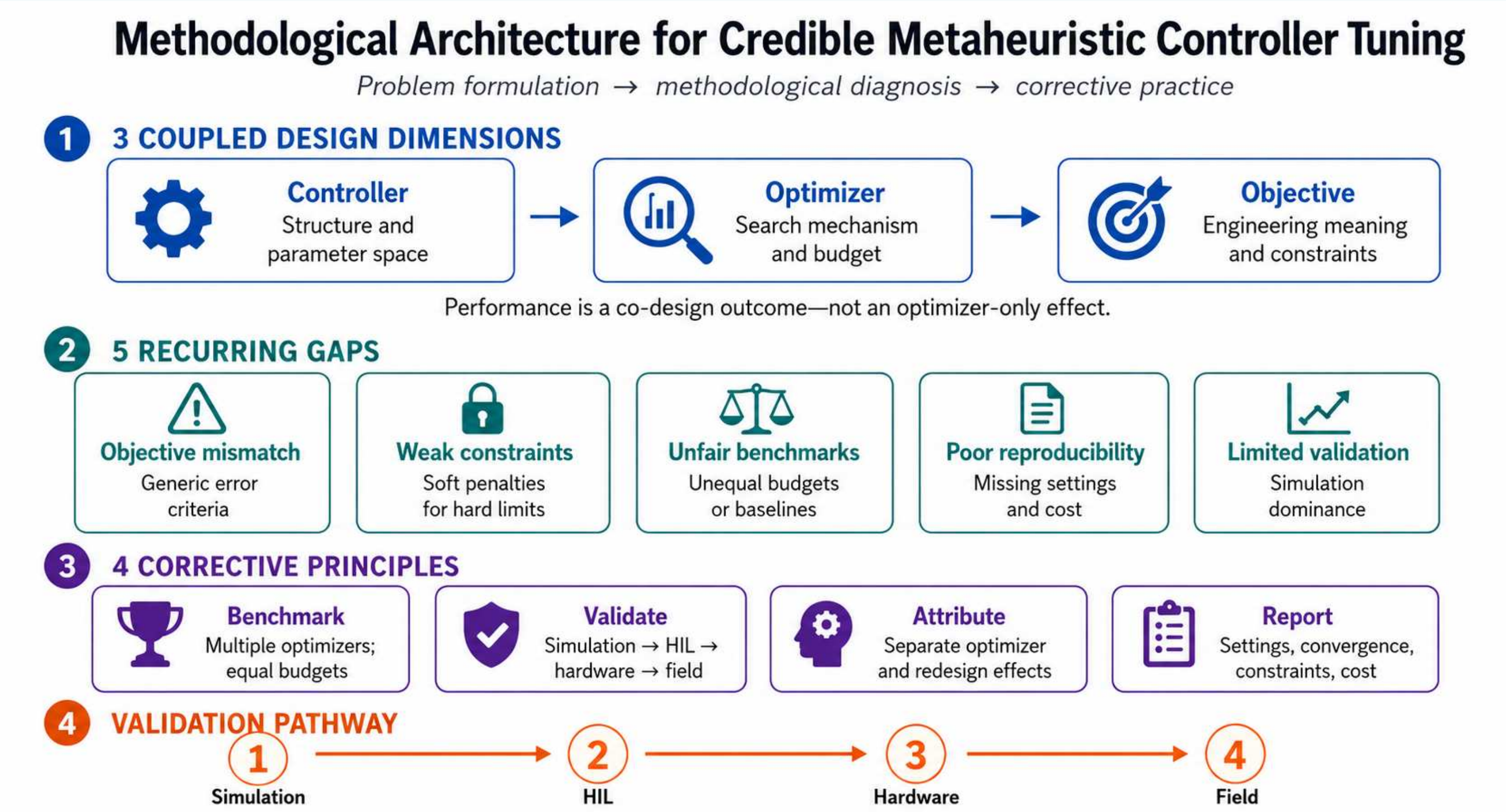
Surrogate-assisted CSTR tuning and frequency-domain ADRC-QFT robust control.



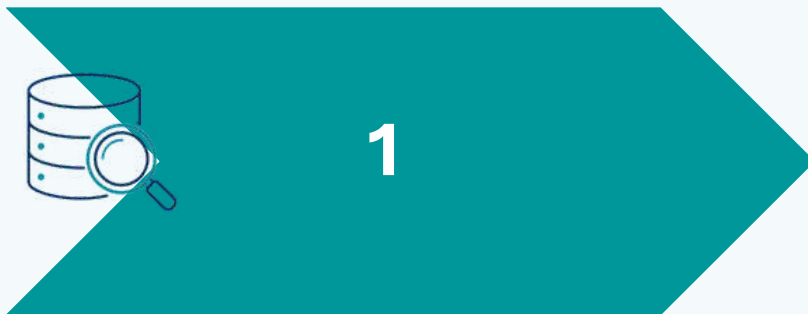
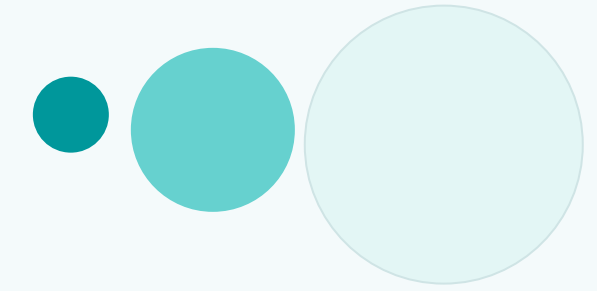
Marine systems

Smith-predictor/fuzzy PID tuning for delay and wave-disturbance rejection.

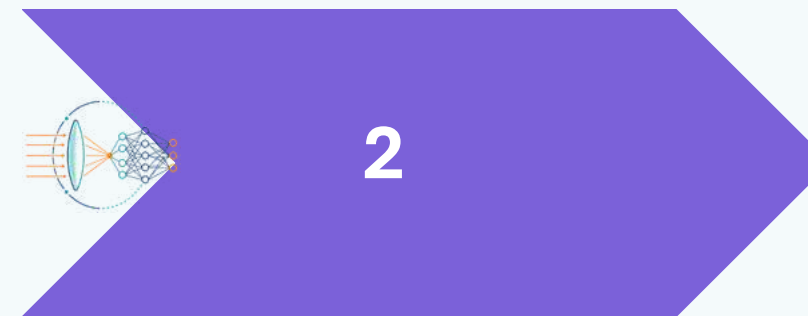
Methodological Gaps Limit Practical Credibility



Conclusion: Formulation Over Algorithm Choice



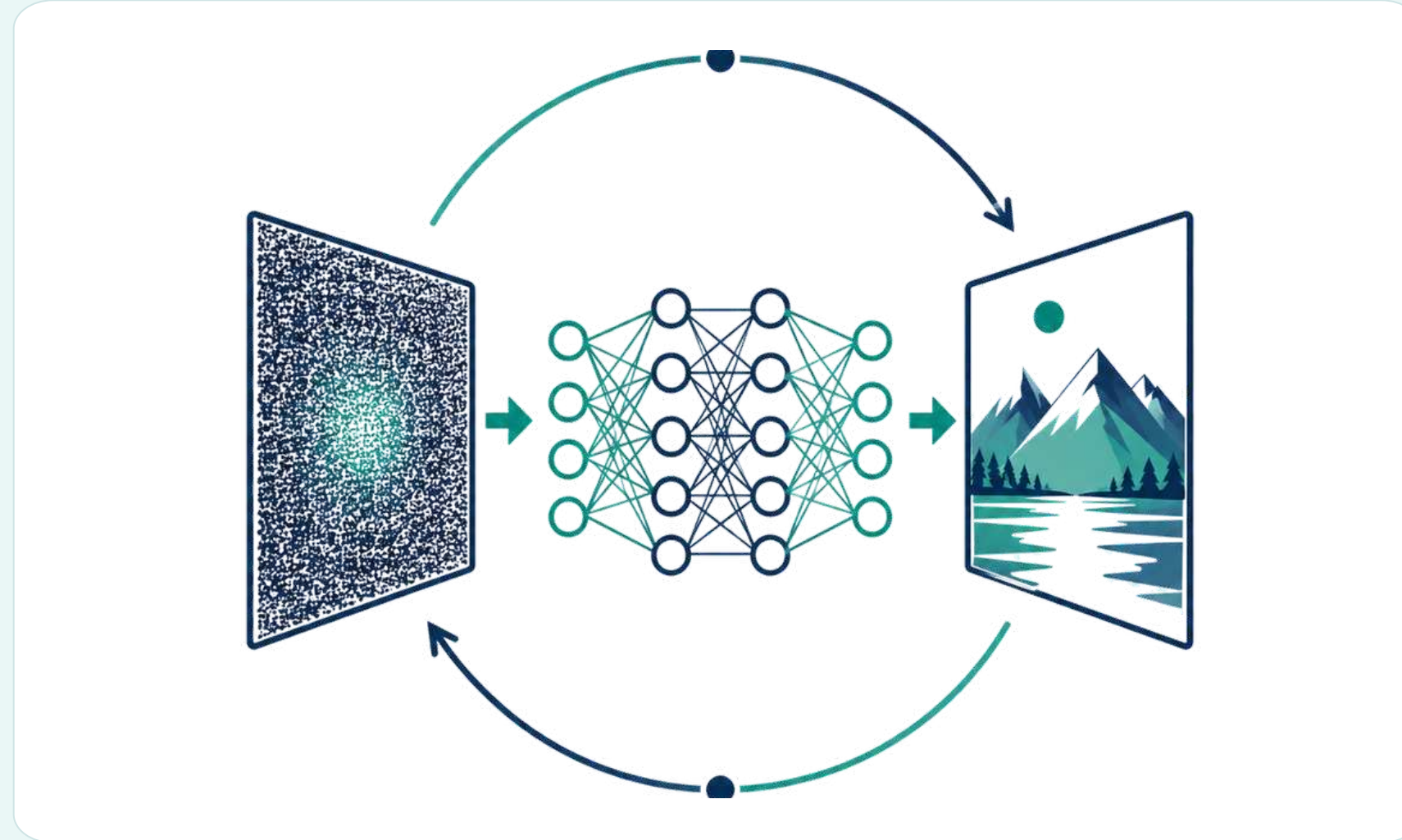
No single optimizer is universally best



Success depends on controller structure, plant dynamics, constraints, and objective design.



Future work must prioritize physical meaning, fair comparison, and reproducibility.



Thank You

Questions & Discussion